

ABHISHEK GOUR

Independent Software Engineer / Systems Builder

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Education

B.Tech, Computer Science & Engineering

2020 – 2024

Prestige Institute of Engineering Management and Research, Indore

Technical Skills

Languages: C++20, Java 21, Python, TypeScript, Kotlin, C

Systems/Frameworks: Spring Boot 3.x, React 19, Next.js 15, Qt6, wlroots, LLVM

Data/Infra: Docker, PostgreSQL, Redis, CMake, Gradle, Git, GitHub Actions

AI/Security: LangChain4j, Google Gemini, TensorFlow Lite, JWT, AES-256, D-Bus, WebSockets

Experience

Freelance Full-Stack Developer

2025 – Present

Remote

- Delivered production platforms for paying clients: Shree Engineers (construction management monorepo) and Comedy Lab (full-stack live comedy booking platform with Spring Boot API, React web client, and native Android app).
- Owned full lifecycle – requirements, architecture, development, deployment – using Spring Boot, Next.js, React, PostgreSQL, and Docker.

Independent Systems Architect

2024 – Present

Self-directed

- Designed and built 14 production-grade artifacts spanning operating systems, compilers, processor architecture, AI orchestration, and security.
- Published a Python library to PyPI with automated CI/CD; submitted a pull request to Google's Closure Compiler with all CI checks passing.

Selected Projects

Tinexus Platform

C++20, wlroots, Qt6, D-Bus

- Architected a Wayland-based Linux desktop environment from scratch, implementing a custom compositor, window manager, and D-Bus system integration layer.
- Achieved sub-150MB idle RAM and sub-100ms launcher latency across the full desktop shell.

DNA Lang & Ribosome Compiler

C++, LLVM, CMake

- Designed a statically typed language with a full compiler pipeline: Lexer, Pratt Parser, AST, custom 3-Address IR, LLVM backend, native x64 codegen.
- Verified correctness with 157 automated tests (100% pass) and 10,000 fuzz runs with zero crashes.

TinexusFlow

Java 21, Spring Boot, LangChain4j, React 19, React Flow

- Built a branching conversation engine that turns linear AI chats into navigable trees, letting users jump between nodes and backtrack to any prior branch.
- Implemented a context-builder service that dynamically reconstructs LLM context per branch, verified end-to-end via a scripted node-creation and resume flow.

Tinu's-QPU

Research, MVL, Formal Proofs

- Formally specified a complete Base-4 processor architecture from logic gates to OS interface, authoring a 10-page paper with theorems proving arithmetic correctness.
- Introduced novel contributions: single-Quat compare flag (QF_CMP), Q-LRU eviction policy, and a 4-level hardware privilege model.

AguAI – Multi-Agent AI Workforce

Spring Boot, LangChain4j, Gemini

- Built a self-orchestrating pipeline where an Architect agent plans a project's file structure in JSON and N concurrently-scoped Codex agents generate code end-to-end.
- Persisted every file-to-agent assignment and live task status in PostgreSQL via a Tracker agent, exposed through a background-threaded REST endpoint.

Smriti – AI-Powered Secrets Manager

Java, Spring Boot, React, Gemini

- Built an enterprise secrets manager with AES-256-GCM encryption, RBAC, and Gemini-embedding semantic search across stored secrets.
- Added Magic Link + Email OTP authentication, full audit logging, and a standalone CLI for environment-variable injection.

Open Source & Achievements

- Google Closure Compiler (PR #4333, open):** Diagnosed and fixed a NullPointerException in InlineObjectLiterals caused by block-scoped functions shadowing hoisted variables; added a regression test, all 6 CI checks passing, CLA signed.
- VoxPulse:** Published a custom wake-word detection library to PyPI (`pip install voxpulse`), MIT-licensed with automated CI/CD.
- 1,089 GitHub contributions in the last year; Pull Shark x2 (two merged open-source PRs); Google CLA signed.
- Certifications:** Fundamentals of Information Security, HTML Web Development Crash Course (Infosys Springboard, 2022).

Last Updated: August 25, 2026